Sayan De Karmakar

+91 9933103342 | sayandekarmakar2020@gmail.com | linkedin.com/in/sayan-de-karmakar

Education

Ramkrishna Mahato Government Engineering College — Purulia, WB
Bachelor of Technology in Electrical Engineering; CGPA: 7.73 — *Aug. 2022 – May 2026*

Kenjakura Molebona High School — Bankura, WB
Senior Secondary (W.B.C.H.S.E); Percentage: 88.60% — *2022*

Kenjakura Molebona High School — Bankura, WB
Secondary (W.B.B.S.E); Percentage: 87.75% — *2020*

Experience & Training

Intern — June 2025 – Aug. 2025
C&S Electric Limited (under Siemens) — *Noida, UP*

- Collaborated with a QA team to monitor production of CONTRACTORS, RELAY, STATAR, SIGNALLING DEVICES & CONTROL PANEL ensuring 100% compliance with IEC standards.
- Attained complete understanding of assembly lines and conducted failure diagnostics and root cause evaluations, interacting with the R&D department.

Advanced Mechatronics Training — Dec. 2024
Siemens SMSCP, GGSF — *Nashik, MH*

- Trained in Siemens TIA Portal (V14) and SIMATIC STEP 7 for PLC programming and control logic implementation.
- Designed pneumatic, electro-pneumatic, and hydraulic control circuits using FluidSIM and Digikit software.
- Applied advanced troubleshooting techniques for automation challenges and real-life diagnostic simulations.

Basic Mechatronics Industrial Automation Training — July 2024
GGSF — *Nashik, MH*

- Studied and implemented pneumatic and hydraulic control systems across 5 industrial automation stations.
- Acquired hands-on experience in Programmable Logic Controllers (PLCs), sensor operations, and automation principles.
- Practiced ladder logic, Sequential Function Charts (SFC), and Data Structure Diagrams (DSD) for integrated systems.

Basic Mechanical and Electrical Training (STA) — Dec. 2024
Siemens — *Airoli, Navi Mumbai*

- Learned the fundamentals of electrical systems, motors & drives, basic electronics, and usage of various electrical measuring instruments.
- Gained hands-on experience with mechanical tools, machining practices, welding techniques, and hydraulic & pneumatic systems.

Projects

Multifunctional Arduino Car | *Arduino UNO, Bluetooth, Sensors*

- Built an Arduino UNO-based robotic car with voice, smartphone, and autonomous control using Bluetooth, ultrasonic sensor, and servo motor.
- Showcased the project in Robo Racing, demonstrating embedded system design and real-time obstacle avoidance.

Real-Time Location Tracking | *ESP32, Arduino Uno*

- Engineered a location tracking system using ESP32 and Arduino Uno for real-time data transmission to a central server.
- Applied wireless communication and system integration techniques in microcontroller programming.

Virtual Computing & Automation | *Raspberry Pi, Python, MobaXterm*

- Set up Raspberry Pi for remote virtual machine access and automation using Python and MobaXterm.
- Gained experience in virtualization, automation, and scripting for IoT-based applications.

Smart Heart Monitoring System | *Biomedical Sensors*

- Designed a wearable device for real-time heart rate monitoring using biomedical sensors and data analysis.
- Aimed to detect irregular patterns and trigger automated alerts for early medical response.

EV Charging & Power Factor Correction | *Single-Phase Boost PFC Converter*

- Developing a single-phase boost Power Factor Correction (PFC) converter to optimize efficient Electric Vehicle (EV) charging.
- Implementing advanced power factor correction techniques to improve power quality and minimize grid-side losses.
- Currently analyzing system performance with the goal of publishing the project findings in an IEEE conference.

Technical Skills

Programming Languages: C(Basic), Arduino(Basic), Python (Basic), C++ (Basic)
Tools & Technologies: MATLAB/Simulink, SolidWorks, Solid Edge, VS Code, Microsoft Office, TIA Portal, PLC Programming, Raspberry Pi, Arduino Uno
Core Areas: Mechatronics (Basic + Advanced), Circuit Design, Power Systems, Electrical Machines, Power Electronics, Instrumentation & troubleshooting
Soft Skills: Leadership, Communication, Teamwork, Problem-solving

Achievements & Certifications

Achievements: Siemens Scholar (Top 200 students from India based on German Dual Education System), WBJEE Qualified (2022)
Certifications: Siemens STA: Basic Mechanical and Electrical Training (2024), Solid Edge (Associate, Professional, Expert), Siemens Mechatronics (Basic + Advanced)
Languages: English, Hindi, Bengali (Native)